

Akhand Pratap Singh

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Summary

Computer Science undergraduate at Lovely Professional University (CGPA: 8.2) with hands-on experience in Machine Learning, NLP, and Generative AI. Built and deployed AI applications using Python, Scikit-learn, XGBoost, FastAPI, Docker, and AWS — including a customer churn prediction pipeline and a RAG-based assistant using LangChain and ChromaDB.

Skills

Languages: Python, SQL

Data Visualization Tools: MS Excel, Power BI

Data Science & ML: Supervised & Unsupervised Learning, Deep Learning, NLP, Scikit-Learn, Pandas, NumPy

Generative AI: Large Language Models, Prompt Engineering, RAG, LangChain, Vector Databases, Embeddings

Backend & MLOps: FastAPI, Django, REST APIs, Docker, AWS (EC2), Git/GitHub

Projects

Conecto – AI-Moderated Social Networking Platform | [GitHub](#) | [Live](#) **Jan' 2026 – Feb' 2026**

- Architected a full-stack social networking platform with profiles, posts, and personalized timelines, supporting 500 concurrent users on a scalable PostgreSQL backend.
- Increased content moderation accuracy to 97% using LinearSVC (NLP) for abusive text detection and NudeNet for NSFW image filtering.
- Developed real-time chat using WebSockets (Django Channels) with secure authentication via Django Auth and Google OAuth.
- **Tech:** Python, Django, PostgreSQL, NLP (LinearSVC), NudeNet, HTML, CSS, JavaScript

Bank Churn Analysis & Customer Retention Modeling | [GitHub](#) **Nov' 2025 – Dec' 2025**

- Designed churn prediction models on 10,000+ customer records using XGBoost and LightGBM, improving ROC-AUC from 0.78 to 0.866 via Optuna tuning.
- Improved business decision-making by building a Power BI dashboard for churn trend visualization and a FastAPI REST service delivering real-time predictions at <10ms latency.
- Deployed a Dockerized ML pipeline on AWS EC2 with model artifacts stored in S3 for scalable inference.
- **Tech:** Python, FastAPI, XGBoost, LightGBM, Scikit-learn, Pandas, NumPy, Docker, AWS, Power BI.

ClassAudit AI: Smart Monitoring & RAG-Based Policy Assistant | [GitHub](#) **Sep' 2025 – Oct' 2025**

- Implemented real-time face recognition using dlib (128-d embeddings, 0.45 threshold), achieving ~87% accuracy with <0.05% false accept rate for secure access control.
- Achieved 90% reduction in manual attendance effort by building a Django backend with role-based access, automated webcam attendance logging, and real-time class monitoring.
- Engineered a RAG-based policy assistant using Groq (LLaMA 3.3 70B), ChromaDB, and all-MiniLM-L6-v2 embeddings, integrated with Chart.js analytics and a 14-point risk scoring model to identify defaulters, classify teacher performance, and export CSV audit reports.
- **Tech:** Python, Django, OpenCV, face_recognition, dlib, Groq API, ChromaDB, sentence-transformers, Chart.js, PostgreSQL.

Achievements & Certifications

- **SAP Certified Data Analyst (2026):** Specialized in augmented analytics and data modeling using SAP Analytics Cloud.
- **Oracle Cloud AI & Data Foundations (2026):** Certified in Generative AI, LLMs, autonomous databases, and cloud data platform architecture.
- **LeetCode Global Top 10% (2025):** Solved **400+** problems across Graphs, DP, Linked Lists, Trees, Arrays.
- **Patent: QR-based IoT Seat Authorization (2026):** Planned a QR-based IoT access control system for secure seat authorization and filed a patent for the innovation.

Education

Lovely Professional University

Bachelor of Technology - Computer Science and Engineering

Aug' 2023 – Present

CGPA: 8.2 / 10